

Mobile Application Development with Flutter

Lecture 50. SQLite Database in Flutter

Introduction

- What is Database?
 - A **database** is a place where we store, manage, and retrieve data in an organized way
- Why Do Apps Need Databases?
 - Apps need databases because data should not be lost when the app closes
- So databases help us:
 - Save data, Read data, Update data, Delete data
 - These four main operations are called CRUD

Types of Databases in Mobile Apps

- In mobile apps, databases are mainly divided into two categories:

1. Local Database
2. Cloud Database

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Local Database

- A **local database** stores data inside the user's mobile device
- Features
 - No internet is required
 - These databases are relatively fast
 - Data is private to a particular device
- Common local databases in Flutter
 - SQLite, Hive, Isar, ObjectBox
- This Lecture
 - In this lecture we focus on SQLite

Real-life examples of SQLite

1. Notes App
2. To-Do List App
3. Expense Tracker App
4. Offline Dictionary App
5. Attendance App for One Teacher

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Cloud Database

- A cloud database stores data on an online server
- Features
 - It usually requires internet
 - Data is saved online
 - Multiple users can access it
 - Data can sync across devices
- Common cloud databases
 - Firebase, Supabase, MongoDB
- For Flutter beginners, Firebase is very popular

Real-life examples of Firebase

1. Chat App
2. Food Delivery App
3. Social Media App
4. Online Quiz App
5. E-commerce App

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SQLite vs Firebase

Feature	SQLite	Firebase
Storage	Inside phone	Online server
Internet required	No	Usually yes
Works offline	Yes	Yes with sync support, but mainly cloud
Multi-user support	No direct support	Yes
Real-time updates	No	Yes
Best for	Offline single-user apps	Online multi-user apps
Complexity	Easy	Medium